

Performance Testing

Date	02 JULY 2026
Team ID	xxxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates the OptiCrop system under different user loads to ensure fast crop recommendation, system stability, and reliable performance during concurrent access.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust (Python), Postman
Type of Testing	Load Testing, Stress Testing, Response Time Testing
Target Module / API	/predict (POST), /recommend (POST)
Test Environment	Local Host (Intel Core i5 / Apple M1, 8GB RAM)
Test Date	July 2, 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal Crop Prediction Requests: Multiple users request crop recommendations simultaneously.	10	30s	Average response time below 300 ms.
2	Continuous User Requests: Simulates farmers submitting requests continuously.	50	60s	Fast and accurate recommendations without failures.
3	High Load Test: Large number of concurrent prediction requests.	200	120s	Stable response with low error rate.
4	Peak Traffic Test: Sudden spike in recommendation requests.	500	60s	System remains responsive and recovers quickly.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 500 ms	220 ms	Pass	Fast ML prediction
2	Response Time (Max)	< 2 sec	760 ms	Pass	Stable under load
3	Throughput (Req/sec)	Run Dependent	165 req/sec	Pass	Handled concurrent requests
4	Error Rate	< 1%	0.0%	Pass	No request failures
5	CPU Utilization	< 80%	38%	Pass	Efficient processing
6	Memory Utilization	< 80%	42% (avg)	Pass	Stable memory usage

Step 4: Observations & Analysis

- **Key Findings:** The OptiCrop system consistently delivered crop recommendations with low response time while maintaining stable performance under normal and high user loads.
- **Bottlenecks Identified:** Minor delay observed during initial model loading before optimization.
- **Optimization Steps Taken:** Machine learning model was loaded once during application startup, reducing prediction time and improving overall system performance.

Step 5: Screenshots / Evidence

